

Noah Khaloo

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Experience

Research Scientist Intern

August 2026–Present

Meta Reality Labs

Redmond, WA

- Develop evaluation pipelines to characterize speech and audio quality from wearable devices operating in noisy, real-world acoustic environments.
- Create benchmark datasets for evaluating enhanced speech and wearable-device audio across representative acoustic conditions.
- Analyze acoustic feature representations to assess how noise suppression, speech enhancement, and target-speaker amplification preserve voice quality.
- Investigate objective and perceptual evaluation methods for enhanced speech, applying DSP and signal-level analysis to wearable-device audio.

PhD Researcher

September 2023–Present

UCSD Phonetics Laboratory

San Diego, CA

- Develop subjective and objective evaluation methods for text-to-speech and zero-shot speaker adaptation.
- Extract signal-level properties of synthetic speech and find correlations with subjective user ratings across settings.
- Develop open-source Python tools for scalable acoustic feature extraction, digital signal processing, and speech analysis.

Machine Learning Engineer (Contract)

June 2026–August 2026

Loreo.ai

Remote

- Evaluated pretrained speaker-diarization models across languages, accents, and noise conditions for real-time transcription.
- Created naturalistic speech benchmarks modeling realistic turn-taking, interruptions, and conversational flow.
- Built reproducible evaluation pipelines, benchmarked GPU/CPU performance, and developed fine-tuning protocols for ASR and diarization models.

AI Trainer (Contract)

June 2024–October 2025

OpenAI

Remote

- Designed research-style prompts across statistics, linguistics, and cognitive science to rigorously test large language model capabilities.
- Validated model outputs for accuracy, completeness, and clarity.
- Developed research-backed rubrics for assessing model performance across prompts.

Open Source tools:

VSPY — Python package that extracts perceptually motivated acoustic features for assessing voice quality and text-to-speech naturalness. Source code: github.com/nkhaloo/vspy.

Selected Publications

Khaloo, N., Holliday, N., & Creel, S. (2026). “Perceptual and acoustic correlates of racial identity in text-to-speech voices. extitInterspeech 2026. Accepted.

Education

University of California, San Diego

September 2023–Present

PhD, Linguistics

San Diego, CA

University of California, Irvine

September 2017–June 2021

Bachelor of Arts, Psychology

Irvine, CA

Technical Skills: Python, PyTorch, R, C++

Core Concepts: Voice-AI, AI evaluation, speech production, acoustic signal processing, human-computer interaction